1 A fairground ride carries passengers in chairs which are attached by metal rods to a rotating central pole, as shown in Fig 1.1. When the pole rotates with angular velocity ω, the rods make an angle θ to the vertical.

For Examiner's Use

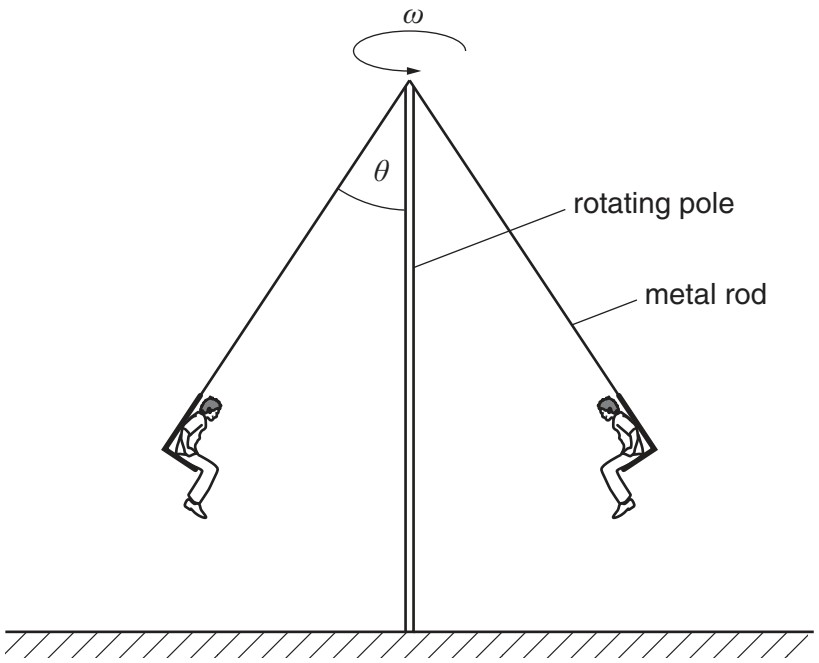

Fig 1.1

It is suggested that $\cos\theta$ is inversely proportional to ω^2.

Design a laboratory experiment, using a small object to represent an occupied chair, to test the relationship between θ and ω. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

[15]

https://xtremepape.rs/